

takeoff. One woman passenger wrote of a London-to-Cape Town flight: "It's a tremendous strain going on and on, being relentlessly woken any time from 3:00am to 4:15am and flying and flying and flying and flying."

Nor was full reliability ever achieved. Many journeys were interrupted by bad weather, and navigational errors remained a source of frequent anxiety to crews, since even the best available radio-navigation devices were fallible. And flying boats often had difficulty in alighting if sea conditions were unfavorable. Safety concerns became acute in 1938, when Pan American lost two of its flying boats. In January, shortly after the opening of a second route to New Zealand, an S-42B exploded in the air near Pago Pago in American Samoa, killing the famous Captain Musick and the rest of its crew. The following July, another Pan American flying boat vanished between Guam and Manila.

Bigger and better

Undeterred by these setbacks, Juan Trippe pressed ahead with introducing a fleet of six Boeing 314s. Twice the size of the Martin 130s, these extraordinary aircraft were the biggest airliners to fly until the age of the jumbo jet, and probably the most luxurious fixed-wing passenger aircraft ever built. They boasted a dining room and seven passenger compartments, one of which, the Deluxe Compartment in the tail, was described in the company's promotional literature as "corresponding roughly to a ship's bridal suite." They carried a maximum of 74 passengers seated, or 40 in sleeping berths. With these magnificent airplanes Pan American was at



IMPERIAL LUXURY

This poster for Imperial's Short flying boat reveals a complex interior layout. Introduced in 1936, the S.23 C-Class, or *Empire*, carried 17 passengers and five crew on stages of long-distance routes to Australia, South Africa, and India.

service by the Americans' refusal to grant landing rights. Furthermore, the pressurized Boeing Stratoliner was pointing the way forward to faster, smoother air travel above the weather. Trippe's next large-scale investment after the 314 was an order for 40 Lockheed Constellations.

The stately elegance of the

flying boats, like the extravagant luxury of the *Graf Zeppelin* and *Hindenburg*, belonged to an era in which long-distance air travel was the preserve of the wealthy, and style seemed a better selling point than functionality. Quickly outmoded, they understandably remain a focus of nostalgic fascination for many modern air passengers who, trapped in the cramped blandness of a contemporary airline interior, like to imagine a time when a journey by air was an experience to be savored and an adventure to be remembered for a lifetime.

last able to initiate a transatlantic passenger service in May 1939, when the *Yankee Clipper* took off from the Marine Terminal in New York for Lisbon and Marseilles.

It was unfortunate for the Boeing 314 that war intervened just as the service to Europe was becoming established. Yet by this time it was in any case already clear, even to Trippe, that the future lay with a new generation of land-planes. Lufthansa's FW 200 had already flown direct from Berlin to New York, and the German airline was only prevented from starting a scheduled

INTERESTING EXPERIMENTS

THE COMPLEX EQUATIONS OF POWER, payload, and fuel often did not add up for flying boats on long-distance routes. One failed attempt at resolving the dilemma of taking off with enough payload to be worthwhile and enough fuel to cross an ocean was the extravagantly powered Dornier Do X of 1929, which had no fewer than 12 engines. One Do X took up 170 passengers, but the airplane was plagued with problems and never entered service. It has been generally dismissed as, in the words of one expert, "an ambitious freak."

Another solution to the power-payload-fuel problem was some form of assisted takeoff. The Germans experimented with catapulting flying boats and floatplanes into the air. In the Short Mayo Composite (below), the British used a flying boat with a light fuel load to lift a heavily loaded seaplane into the air, which was then released to continue its flight. The idea worked – from 1938–39, the Mercury made a number of long-distance mail flights – but was not very practical and remained an aviation curiosity.

COMPOSITE SOLUTION

Invented by the technical manager of Imperial Airways, Major R. Mayo, the Composite flying boat – first flown in July 1938 – involved a Short S.20 seaplane, Mercury, riding piggyback on Maia, a Short S.21 flying boat.

Mother-plane Maia carries Mercury to cruising height



TUCKING IN

The Pan American Boeing 314 *Yankee Clipper*, which flew the first scheduled transatlantic service in 1939, was the most spacious and luxurious flying boat of all. Mealtimes in the dining room featured full waiter service and the finest haute cuisine.

